

Hardware Assessment Report

Gen Mot Coils

Scan Date: May 27, 2026 at 07:15 AM



Estimated Lifespan: 1-2 years

System Information

Hostname: DesignSw

OS: Microsoft Windows 10 Pro 10

CPU: 12th Gen Intel(R) Core(TM) i7-12700

RAM: 15.7 GB

Overall System Health: 55/100 (Grade C)

Estimated Remaining Lifespan: 1-2 years

Critical Issues Detected:

- GPU: D (34/100) — Replace soon, degraded performance

Component Health Summary

Component	Score	Grade	Status
12th Gen Intel(R) Core(TM) i7-12700	80	A	No action needed
15.7 GB (2 modules)	80	A	No action needed
WD Blue SN570 500GB	70	B	No action needed
PC801 NVMe SK hynix 512GB	70	B	No action needed
NVIDIA T400 4GB	34	D	Replace soon, degraded performance
Intel(R) UHD Graphics 770	67	B	No action needed
Dell Inc. 0N0992	81	A	No action needed

CPU — 12th Gen Intel(R) Core(TM) i7-12700

Grade A (80/100)

Part/Serial: BFEBFBFF00090672 Status: ■ OK

→ Reapply thermal paste to reduce operating temperatures

→ Clean CPU heatsink and fan to prevent thermal throttling

RAM — 15.7 GB (2 modules)

Grade A (80/100)

Part/Serial: M323R1GB4BB0-CQKOD Status: ■ OK

→ Run MemTest86 to verify memory stability and rule out bad sectors

→ Ensure RAM is configured in dual-channel mode for optimal bandwidth

Storage — WD Blue SN570 500GB

Grade B (70/100)

Part/Serial: E823_8FA6_BF53_0001_001B_448B_4E9F_..Status: ■ OK

→ Enable TRIM support in the operating system for optimal endurance

→ Avoid filling the SSD beyond 80% capacity to extend lifespan

Storage — PC801 NVMe SK hynix 512GB

Grade B (70/100)

Part/Serial: ACE4_2E00_2589_1211_2EE4_AC00_0000_..Status: ■ OK

→ Enable TRIM support in the operating system for optimal endurance

→ Avoid filling the SSD beyond 80% capacity to extend lifespan

GPU — NVIDIA T400 4GB

Grade D (34/100)

Part/Serial: NVIDIA T400 4GB Status: ■ REPLACE_SOON

→ Update GPU drivers to the latest stable version

→ Clean GPU fans and heatsink to prevent thermal throttling

GPU — Intel(R) UHD Graphics 770

Grade B (67/100)

Part/Serial: Intel(R) UHD Graphics Family Status: ■ OK

→ Integrated GPU is soldered to the CPU/motherboard — cannot be upgraded separately

→ To improve GPU performance, a motherboard + CPU replacement is required

Motherboard — Dell Inc. 0N0992

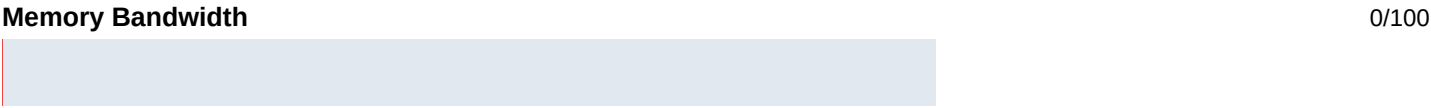
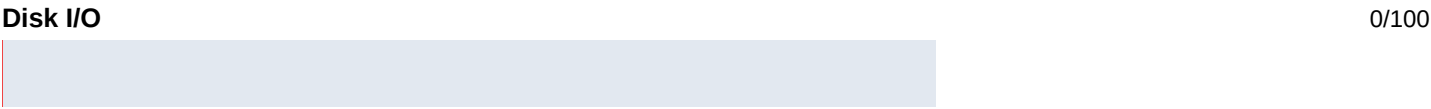
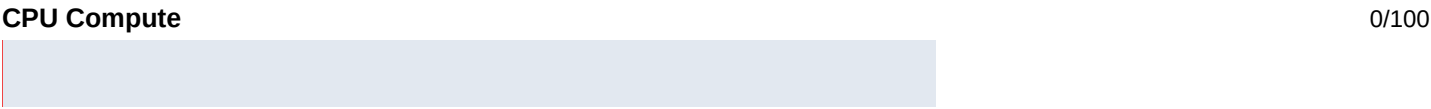
Grade A (81/100)

Part/Serial: /H3PFVQ3/CNFCW00268001J/ Status: ■ OK

→ Update BIOS firmware to the latest version for compatibility fixes

→ Visually inspect capacitors for bulging or leaking (sign of failure)

Overall Benchmark Score: 0/100



Category	Status	Bottlenecks
Web Development	Fully Supported	None
Desktop Development	Minimum Requirements Only	Recommended 16GB RAM (has 15.7GB)
Mobile Development	Minimum Requirements Only	Recommended 16GB RAM (has 15.7GB)
Databases	Minimum Requirements Only	Recommended 16GB RAM (has 15.7GB)
Virtualization	Minimum Requirements Only	Recommended 32GB RAM (has 15.7GB)
AI & Data Science	Not Supported	Recommended 32GB RAM (has 15.7GB); Needs 4096MB VRAM (has...
CAD & Engineering	Minimum Requirements Only	Recommended 32GB RAM (has 15.7GB); Recommended 4096MB VRA...
Game Development	Not Supported	Recommended 32GB RAM (has 15.7GB); Needs 4096MB VRAM (has...
Video Editing	Minimum Requirements Only	Recommended 32GB RAM (has 15.7GB); Recommended 6144MB VRA...
Basic Office & Browsing	Fully Supported	None

Total Recommendations: 5

- [MEDIUM] GPU
Replace soon, degraded performance
- [HIGH] System
Recommended 32GB RAM (has 15.7GB)
- [HIGH] System
Needs 4096MB VRAM (has 2048.0MB)

Total Estimated Upgrade Cost

Rs. 24,000 INR (~\$265 USD)

Prices as of 2026-04-07 | Sources: Amazon.in, MDComputers, PrimeABGB

Priority	Component	Current	Recommended Upgrade	■ INR	\$ USD
OK	RAM	15.7GB DDR5	DDR5 16GB 5600MHz	Rs. 6,000	\$65
OK	CPU	12th Gen Intel(R) Core	Intel i5 (12th/13th Gen)	Rs. 18,000	\$200
			TOTAL	Rs. 24,000	\$265

Upgrade Justification

- RAM: Upgrade from 15.7GB to 16GB DDR5
Source: PrimeABGB (<https://www.primeabgb.com/ddr5-ram>)
- CPU: Upgrade from 12-core/2100.0MHz to modern mid-range CPU
Source: MDComputers (<https://mdcomputers.in/processors?brand=intel&tier=i5>)

* Prices are approximate and may vary. Consult your vendor for exact quotes. Currency rate: 1 USD = ■92.0